

A TRIP TO HAWAII



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Student Workbook

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UCI MATH CEO Curriculum Module

University of California, Irvine

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About UCI MATH CEO



UC Irvine Math CEO is a free, innovative comprehensive, high-quality math enrichment program designed to empower secondary students in underserved communities in Southern California through joyful after-school activities, collaboration, mentorship and intergenerational learning.



MATH CEO Website (sites.ps.uci.edu/mathceo/)



Math CEO YouTube Channel (youtube.com/channel/UCF_S03ELBGZTpX8DmX4ZRgA)

Our Goals.

- | | |
|----------------|---|
| Promote | students' interest and proficiency in mathematics, through engaging mathematical explorations that are relevant to youth, and inclusive, culturally-responsive, informal pedagogical practices. |
| Support | students' socio-emotional development and wellbeing, by fostering positive relationships with peers and college mentors. |
| Foster | a college-going identity in youth, increase their familiarity with college and set the foundations to be successful in STEM careers. |

Special Thanks.



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Introduction



Let's learn about traditional lengths in Hawaiian Culture and use our bodies to do math and discover patterns!

Overview.

1. *Hawaiian Measures (85 min).*

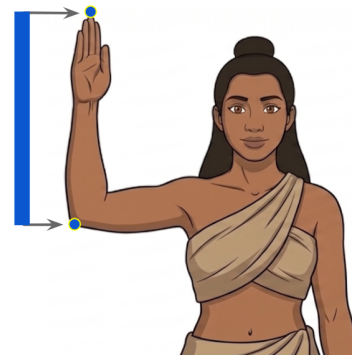
Let's learn about non-standard measures from Hawaiian natives, and compare their proportions.

2. *Equivalencies (40 min).*

Let's build a table of equivalencies of measures, investigate, and draw conclusions!

3. *Reflections (10 min).*

Let's look back on what we learned and reflect on measuring across cultures.



Math Topics.

- Conventional and non-conventional units of length
- Estimations
- Ratios
- Unit ratios
- Equivalencies

Goals: Enrichment

- Learn about measures in different cultures.
- Measure things in different ways.

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PART 1

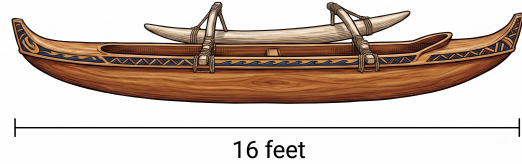
Hawaiian Measures (85 min)



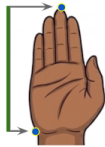
Measuring With Our Bodies

In Hawaiian culture, the body was commonly used to measure things like the length of a canoe, the height of a person, or the width of a taro patch. Using body lengths was a convenient way for the Hawaiians to estimate distances without instruments.

Imagine we are in the middle of the ocean without any rulers or measuring tapes. We want to know how long our canoe is in order to arrange our cargo.



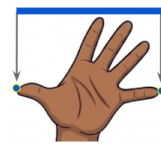
pī'ā



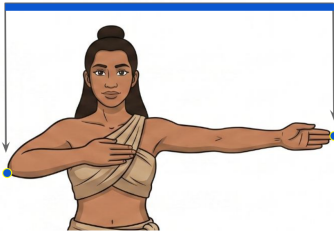
poho



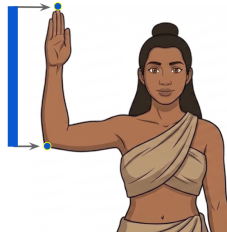
kīko'o



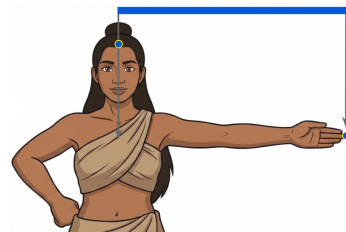
muku



ha'ilima



iwilei



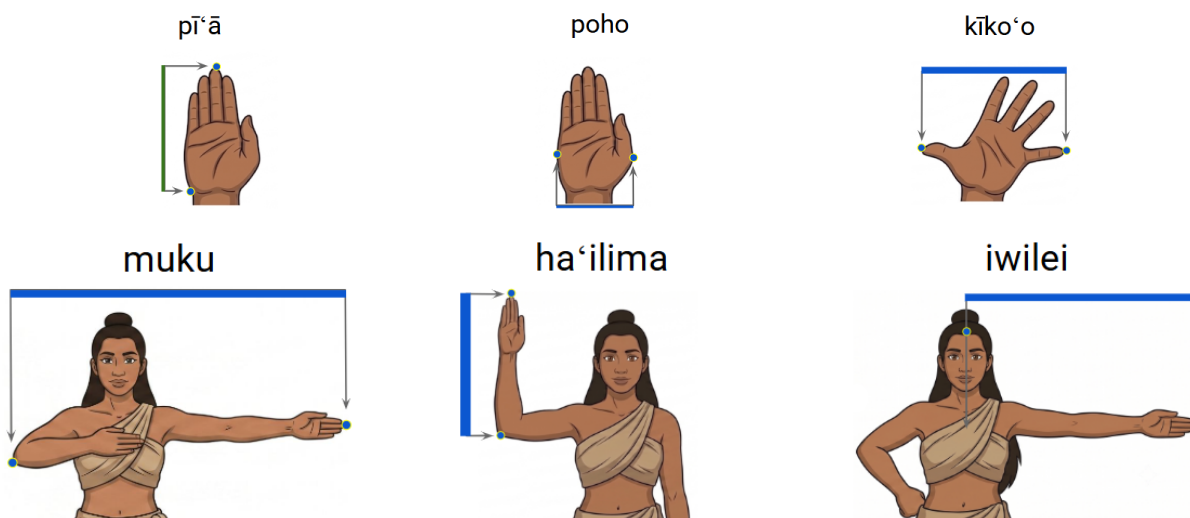
1. Using the Hawaiian measures, what measurement could we use to most efficiently measure this canoe?

2. Estimate: About *how many* of that measurement would be equal to these 16 feet?

3. Discuss.

- What other things do you think the iwilei/ha'ilima/muku were used to measure?
- How might you use these measurements in your daily life?

Lengths!



1. Notebook length.

Measure the **long side** of a notebook in poho, and then in kīko'o. What did you get approximately?

- ____ poho
- ____ kīko'o

2. Fill in the blanks:

- 1 kīko'o is about the same as ____ poho.
- 4 poho is about the same as ____ iwilei.

3. Create more comparisons:

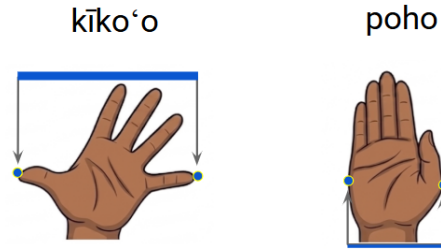
- 1 _____ is about the same as _____
- _____ is about the same as _____

4. Discuss.

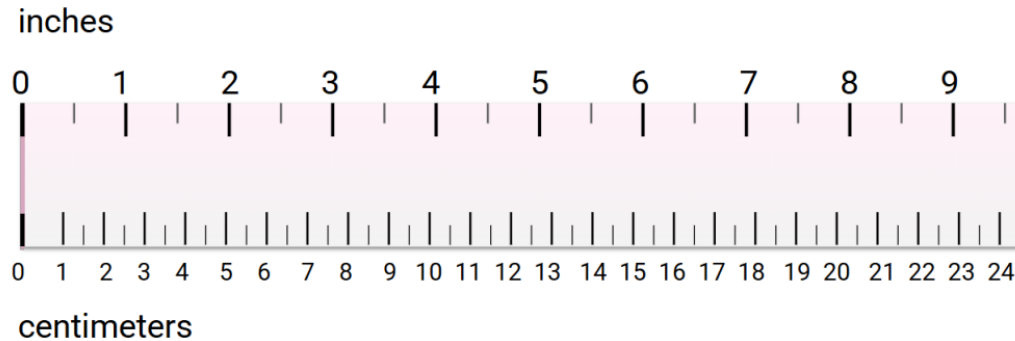
1. Why/when do you think these units of measure were invented?
2. What are the advantages of these units? And disadvantages?
3. Do you use your body or objects to take measures? How?

Let's Build a Measuring Tape

We will first analyze a measuring tape that relates inches and centimeters. We will then build a measuring tape that relates *kīko'o* and *poho*.



1. Use the tape with marks for *in* and *cm* to answer the questions.



1. If an object is 4.5 in long, how many cm is that? _____
2. 6.35 cm is how many inches? _____

2. Now build a single measuring tape for *kīko'o* and *poho*.

Key principles for building a measuring tape:

- We start at zero (0).
- Numbers go from left to right.
- Label each time to identify the measure being marked.

Instructions:

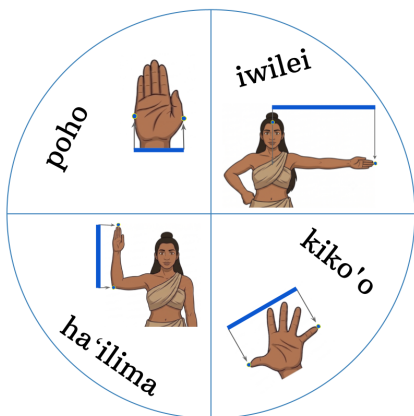
- Use the roll (one side) and markers.
- Whose hands will be used?
- Use an edge for *kīko'o*, the other for *poho*.
- Draw a 0 mark for both units. Then: 1, 2, 3, ...
- No gaps! No overlaps!



Answer the following questions:

- (a) If an object is 4.5 *kīko'o* long, how many *poho* is that? _____
- (b) 5 *poho* is equal to how many *kīko'o*? _____

Spin It!



- Spin the spinner twice to get two units (unit#1, unit#2).
- Roll the die. Say, you got p .
- Create an equation:

$$[p] \cdot \text{unit\#1} = [?] \cdot \text{unit\#2}.$$

Then, solve it. (You can approximate a bit!)

1. Round 1

2. Round 2

3. Round 3



Our Data

In your group, complete the following data. Measure in *inches*. Round to the nearest quarter of an inch.

Person	iwilei	kīko'o	iwilei / kīko'o
1)			
2)			
3)			
4)			
5)			
6)			
7)			
8)			
9)			
10)			
a possible center			

1. How much variation is there in the measures?

2. What seems to be the **center** (a representative value) of each measure?

3. Are there any **outliers** (data far from the typical values)?

4. Now use a calculator to find the last column in the table. Round to 1 decimal place. What do you notice?

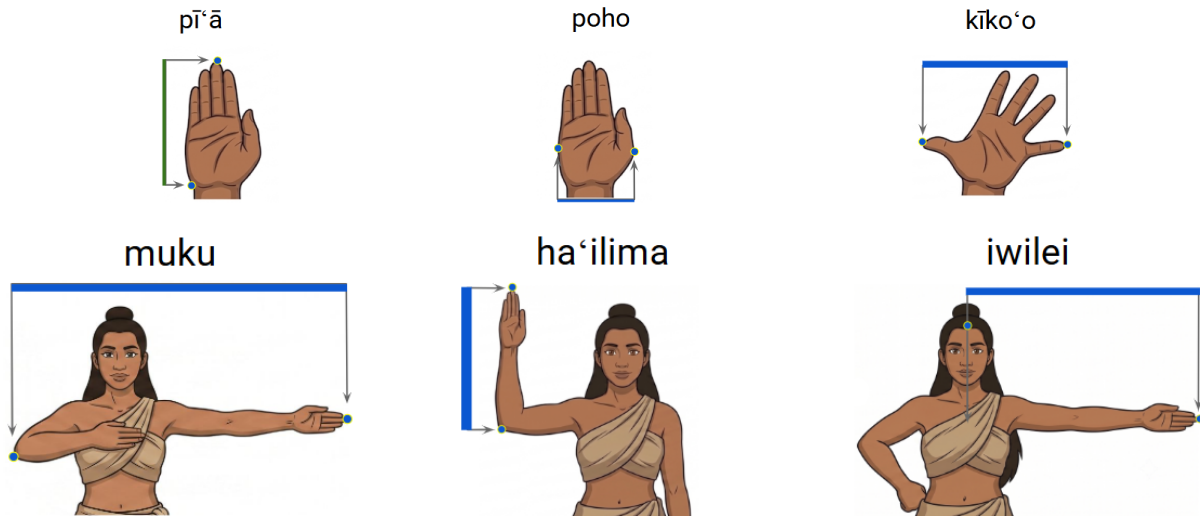
PART 2

Equivalencies (40 min)



Now let's connect the measures, build a table of equivalencies, hunt for patterns, and discover new units using our body.

Equivalencies



1. Complete the table of equivalencies.

	poho.	kīko'o.	ha'ilima.	iwilei.
1 poho =			1/4	
1 kīko'o =	2			
1 ha'ilima =				
1 iwilei =				

2. Which strategies did you use to fill the table? What was the most challenging part?

3. What do you notice about the table you filled? Write all things that you notice!

Use your body!

Come up with a body measure
— pose (describe it, draw it, and
show it) equivalent to:

It's ok to get close equivalencies!



1. 4 ha'ilima

2. 3 ha'ilima

3. 4.5 kīko'o

4. $\frac{2}{5}$ ha'ilima

5. $3.\bar{3}$ iwilei

Pele

In Hawaiian religion, Pele is the goddess of volcanoes and fire and the creator of the Hawaiian Islands. Often referred to as “Madame Pele” or “Tūtū Pele” as a sign of respect, she is a well-known deity within Hawaiian mythology and is notable for her contemporary presence and cultural influence as an enduring figure from ancient Hawaii.

Legend told that Pele herself journeyed on her canoe from the island of Tahiti to Hawai’i. When on her journey, it was said she tried to create her fires on different islands, but her sister, Nāmaka, was chasing her, wanting to put an end to her. In the end, the two sisters fought each other and Pele was killed. With this happening, her body was destroyed but her spirit lives in Halema’uma’u on Kīlauea. They say, “Her body is the lava and steam that comes from the volcano. She can also change form, appearing as a white dog, old woman, or beautiful young woman.”

—Adapted from Wikipedia ([en.wikipedia.org/wiki/Pele_\(deity\)](https://en.wikipedia.org/wiki/Pele_(deity)))

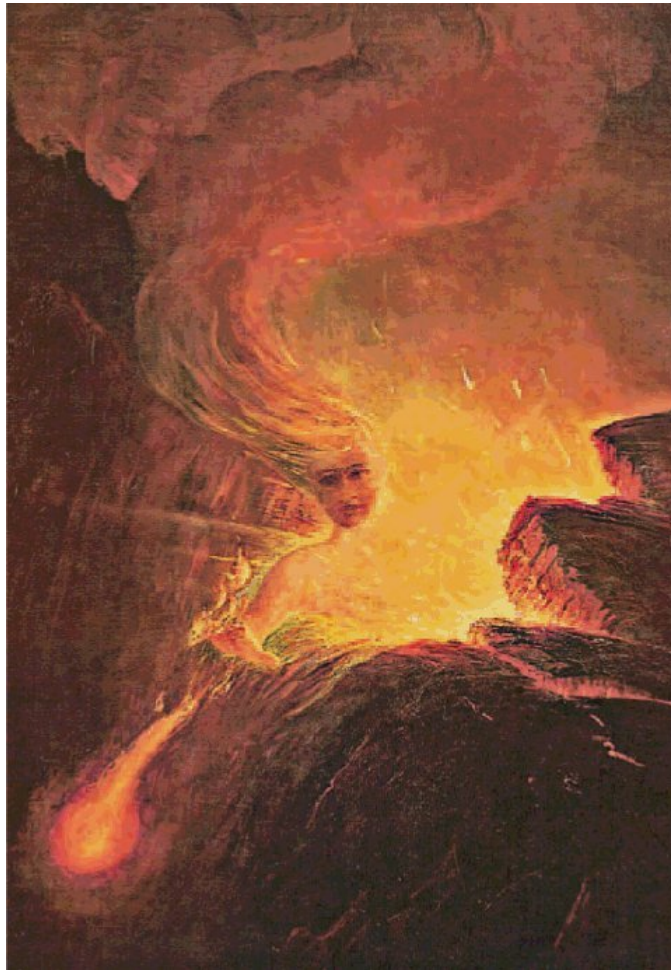


Figure 2.0.1 Pele, by D. Howard Hitchcock (c. 1929)

Pele

1. Sculpture of Pele.

You want to make a sculpture of Pele. You want it to be between 45 and 50 feet tall. Complete the table of ranges with the Hawaiian measures *relative* to the statue.

You can think of it as follows: if a giant person is 45–50 ft tall, what is the length, in ft, of 1 poho of that person?

Space for calculations:

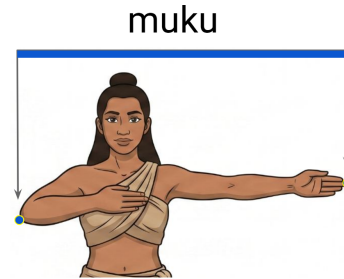
unit	how long? (feet)
1 poho =	
1 kīko'o =	
1 ha'ilima =	
1 iwilei =	
height =	45 to 50

2. A new Pele.

You want to make a small figure of Pele. The length of 1 kīko'o of this figure is 1.25 inches. What should be the height of the figure, in inches?

The Muku

The muku is a traditional Hawaiian unit of length. It is equivalent to the length of a stretched arm, from the tip of the middle finger to the opposite elbow. In other words, it is the distance between the fingertips of one hand when the arm is fully extended, and the elbow. The muku is a very personal and practical measure, as it is based on the individual's own body proportions.



In Hawaiian culture, the muku was commonly used to measure things like the length of a canoe, the height of a person, or the width of a taro patch. It was a convenient way to estimate distances without instruments. This measurement system was passed down through generations, allowing Hawaiians to communicate and understand measurements in their lives.



Comparing the muku to other cultures, we find similar concepts in different parts of the world. For example, the ancient Egyptians used the cubit (the length from the elbow to the tip of the middle finger). Like the muku, the cubit was a personal measure that varied slightly from person to person. The ancient Romans used the pace, which was the distance covered by two steps. These measurements were also based on the human body and were used for practical purposes.

The cultural relevance of the muku goes beyond measurement. It reflects the close connection Hawaiians had with their environment and the importance of human interaction in their lives. By using a measure based on their own bodies, Hawaiians emphasized the relationship between people and the world around them. It also highlights the resourcefulness and ingenuity of the Hawaiian people in finding practical solutions to their needs.

Today, while the muku is not widely used in modern Hawaii, it still holds cultural significance. It serves as a reminder of the rich heritage and traditions of the Hawaiian people. It is also a symbol of the deep connection Hawaiians have with their land, history, and ancestors.



Figure 2.0.2 *Taro patch*, a plant whose root when made into poi forms the principal food of the natives of Hawaiian Islands.

PART 3

Reflections (10 min)



Reflecting About Today

1. What did you learn about how people may solve mathematical problems differently?

2. Do you measure things using body parts? What are the advantages and disadvantages of this way of measuring?

3. What is a **unit ratio**? Why is it helpful?

4. What did you learn about Hawaii today?

Further Reflections

1. How might the environment and cultural practices of a community influence the creation of distinct measures?
2. Share your thoughts on the advantages and disadvantages of traditional units compared to modern standardized measurements.
3. Have you ever encountered unique measures in your own cultural background or learned unconventional ways of measuring?
4. Since bodies vary in height and proportions, do you think there were preferred body measurements in Hawaiian units of measures OR was variation welcomed? (i.e., an 8 year old's iwilei is different from a 23 years old's iwilei, how was this possibly taken into account?)
5. How do you think that different cultures around the world developed their own systems of measurement? What factors might have influenced the choice of units in these systems?
6. Do you think it's important to preserve traditional units of measurement, even though we now have standardized systems like the metric system? Why or why not?
7. How might learning about traditional units of measurement from different cultures broaden our understanding of the diversity of human knowledge and experience?
8. If you were to create your own system of measurement based on your body or objects around you, what units would you use and why?
9. Every country has different measuring systems. Did you know that there is a unit of measurement that was used in ancient times and is based on the time it takes to cook rice? Can you guess which culture used this unique method of measuring time and what the unit is called?
10. How else do you think we can use body measurements to approximate size, distance, etc. in our everyday lives?
11. What are the benefits of these units of measurements? What are the consequences? How do these units of measurements relate to non-standard units from other societies?

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